

ATHARVA WAGH

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SUMMARY

Computer Science graduate student with 2+ years of experience building scalable software systems, backend services, and distributed applications using Java, C#, Python, and C/C++. Strong foundation in data structures, algorithms, and software design principles with hands-on experience developing, testing, debugging, and deploying production-ready systems. Experienced working in Linux environments, containerized infrastructures, and collaborative engineering teams using Git-based workflows. Passionate about building reliable, high-performance software systems that enhance real-world operational awareness and product innovation.

TECHNICAL SKILLS

Languages: Java, Python, C++, C#, JavaScript, SQL, TypeScript

Web & Backend Development: RESTful APIs, FastAPI, Node.js, Spring Boot, .NET, React

Machine Learning & LLM Systems: PyTorch, TensorFlow, Scikit-learn, LangChain, RAG Pipelines, Multi-Agent Systems, Model Evaluation, Statistical Modeling

Databases: PostgreSQL, MongoDB, MySQL, Firebase, Cassandra, AWS Redshift, SQL Server

Devops & Cloud: Git, GitHub, CI/CD, Docker, Kubernetes, Linux, Terraform, AWS (EC2, S3, IAM, SageMaker)

Data & Distributed Systems: Kafka, PySpark, Parallel Processing, ETL Pipelines

Testing & Engineering Practices: Unit Testing, Debugging, Code Reviews, Secure Coding Practices, SDLC

Machine Learning & Analytics: Scikit-learn, TensorFlow, PyTorch, Model Evaluation

EDUCATION

– **University of Southern California, Los Angeles, CA**

Master of Science in Computer Science

GPA: 3.3/4.0

Aug 2025 - May 2027

– **University of Mumbai, Mumbai, India**

Bachelor of Technology in Information Technology (Blockchain Honors)

GPA: 4.0/4.0

Aug 2020 - May 2024

PROFESSIONAL EXPERIENCE

– **Software Developer [Full Time]**

Mumbai, India

Institute of Management and Foreign Studies (IMFS)

Jul 2024 - May 2025

- * Designed, analyzed, and developed backend software solutions using Python and .NET APIs to support enterprise CRM and transactional systems.
- * Built and deployed containerized microservices using Docker and Kubernetes, improving system response time by 83% and enabling scalable, reliable deployments.
- * Developed RESTful APIs for structured data ingestion, transformation, and database integration using MySQL and PostgreSQL.
- * Implemented automated data processing workflows and analytics modules using AWS SageMaker to support business logic and performance insights.
- * Implemented ONNX cross platform machine learning model deployment to develop models in PyTorch and deploy in .NET mvc for scalability and production performance

– **Web Developer Intern [Co-Op]**

Mumbai, India

Mabella SkinCare

Jun 2023 - May 2024

- * Engineered a full-stack enterprise CRM system using React, TypeScript, Flask, and PostgreSQL, supporting consistent transactional workloads.
- * Designed optimized database schemas and caching methods to support consistent high-load operations.
- * Designed & Deployed a OCR based product recommendation system using Tesseract, Fast API & Docker

ACADEMIC PROJECTS

– **LLM Powered Interview Officer**

2024

LLM Agents | LangChain | Web Scraping | RAG | Docker | WebSockets | NLP

- * Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain to improve contextual reasoning and response accuracy.
- * Developed a React-based UI with WebSocket-driven real-time interaction for low-latency workflows.
- * Iteratively evaluated and optimized LLM performance under real-time interaction constraints, leveraging AI-assisted development workflows to accelerate feature delivery.

– **TCR Sequence Extraction from RNA-seq Data**

2025

Bioinformatics Data Pipeline | Python | R | RStudio

- * Achieved 76% alignment/extraction efficiency in validation runs; implemented cross-tool comparison and overlap visualization to quantify concordance between extraction tools (MiXCR vs TRUST4).
- * Built reproducible notebooks and automated data-download scripts (ENA), caching layers to avoid redundant computation, and end-to-end reporting for downstream analysis and clinical interpretation.

– **Predictive Analytics for College Admissions Application**

2023

Machine Learning Project | Python | TypeScript | AWS Sagemaker | NumPy | Sci-Kit Learn | Node.js | Seaborn

- * Collected application data; performed EDA, handled outliers, and applied mean-based imputation; built ensemble pipelines with Random Forest and XGBoost for multi-class classification
- * Achieved ROC AUC scores of 0.81 (train) and 0.78 (test); deployed model via FastAPI with REST endpoints for real-time predictions

RESEARCH EXPERIENCE

– **Bridging Energy Gaps: Blockchain-Enabled P2P Trading for Renewable Energy**

2024